

Week 3: Building and Tuning an AI Model

Introduction

The objective of Week 3 was to develop, train, evaluate, and optimize a machine learning model capable of predicting student academic outcomes using educational engagement and assessment data. This phase builds upon the preprocessing and feature engineering work completed during Week 2 and represents the core predictive component of the LearnPulse-AI project.

LearnPulse-AI is an Early Warning System designed to identify patterns associated with student success, failure, distinction, and withdrawal. By analyzing student interactions with online learning resources, assessment performance, demographic information, and academic history, the system aims to support proactive interventions that improve student retention and academic achievement.

The primary goal of this task was to construct a predictive model that can classify student outcomes while also identifying the most influential factors affecting academic performance.

Dataset Description

The dataset used in this project was derived from the Open University Learning Analytics Dataset (OULAD), a publicly available educational dataset widely used for learning analytics research.

The processed dataset contained:

- 32,593 student records
- 15 predictive features
- Academic outcome labels for each student

The target variable was the final academic outcome of the student.

Possible outcomes included:

- Pass
- Fail
- Distinction
- Withdrawn

The dataset incorporated information related to:

- Student demographics
- Educational background
- Previous attempts
- Study credits
- Assessment performance
- Virtual Learning Environment (VLE) activity

This combination of academic and engagement-related information provides a strong foundation for predictive modeling.

Data Preprocessing Summary

Before training the model, the dataset underwent extensive preprocessing and feature engineering.

The preprocessing workflow included:

Missing Value Handling

Missing values were identified and addressed appropriately. Missing values in the IMD Band column were replaced using suitable techniques to ensure dataset completeness.

Data Cleaning

Irrelevant inconsistencies were removed and all records were standardized into a consistent format suitable for machine learning algorithms.

Feature Engineering

Several new features were created using assessment and engagement data.

Engineered features included:

- Average Assessment Score
- Assessment Count
- Total Clicks
- Average Clicks

These features were designed to capture student engagement and academic performance more effectively than raw records.

Encoding

Categorical variables such as gender, region, education level, age group, and final result were converted into numerical values using label encoding.

Normalization

Numerical features were normalized using MinMax Scaling to ensure that variables with larger numerical ranges did not dominate the learning process.

After preprocessing, the final dataset contained no missing values and was fully prepared for machine learning model development.

Algorithm Selection

A Random Forest Classifier was selected for this project.

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve predictive performance and reduce overfitting.

The algorithm was selected because:

- It handles categorical and numerical features effectively.
- It performs well on structured datasets.
- It is robust against overfitting.
- It provides feature importance scores.
- It generally requires less tuning compared to many advanced algorithms.

Since the LearnPulse-AI dataset contains a mixture of demographic, behavioral, and assessment-related information, Random Forest was considered an appropriate choice.

Model Training

The processed dataset was divided into training and testing sets.

The feature matrix consisted of fifteen predictive variables, while the target variable represented the final student outcome.

The dataset dimensions were:

- Features Dataset Shape: (32593, 15)
- Target Dataset Shape: (32593)

The training set was used to learn patterns from historical student records, while the testing set was reserved for evaluating model performance on unseen data.

A baseline Random Forest model was trained using default parameters.

Baseline Model Evaluation

The initial Random Forest model achieved an accuracy of:

Accuracy = 68.05%

The classification report revealed varying levels of performance across different outcome categories.

The model performed strongest when predicting students who passed or achieved distinction.

Performance was comparatively lower for students who failed or withdrew, which is expected because these categories often exhibit more complex behavioral patterns and class imbalance.

Although the baseline model demonstrated reasonable predictive capability, additional tuning was performed to further optimize performance.

Hyperparameter Tuning

To improve model performance, GridSearchCV was used to systematically evaluate multiple combinations of Random Forest hyperparameters.

The following parameters were explored:

- Number of Trees (n_estimators)
- Maximum Tree Depth (max_depth)

After evaluation, the optimal configuration was identified as:

Best Parameters

- max_depth = 20
- n_estimators = 200

Best Cross-Validation Accuracy

68.64%

Hyperparameter tuning allowed the model to explore more complex decision boundaries while maintaining generalization performance.

Final Model Evaluation

Using the optimized hyperparameters, the model was retrained and evaluated on the test dataset.

Final Accuracy

68.26%

The final classification report showed balanced performance across multiple outcome categories.

Overall Metrics:

- Accuracy: 68.26%
- Macro Average F1 Score: 0.62
- Weighted Average F1 Score: 0.66

These results indicate that the model successfully learned meaningful relationships between engagement behavior, assessment activity, and academic outcomes.

Feature Importance Analysis

One of the advantages of Random Forest is its ability to estimate feature importance.

The most influential features identified by the model were:

Feature	Importance
Assessment Count	19.93%
Average Score	18.20%
Total Clicks	13.05%
Average Clicks	10.15%
Student ID	8.62%
Region	5.42%
Code Module	5.25%

These findings indicate that student engagement and assessment activity are the strongest predictors of academic outcomes.

Students who completed more assessments and interacted more frequently with learning resources generally exhibited better academic performance.

This aligns with educational research and supports the goals of the LearnPulse-AI Early Warning System.

Challenges Encountered

Several challenges were encountered during model development.

The original dataset contained over ten million VLE interaction records, requiring careful aggregation before use.

Missing values needed to be addressed before model training.

Feature engineering required merging information from multiple dataset sources and generating meaningful engagement metrics.

Hyperparameter tuning also increased computational requirements due to repeated model training during cross-validation.

Despite these challenges, a complete end-to-end machine learning pipeline was successfully implemented.

Conclusion

This phase of the LearnPulse-AI project successfully developed and optimized a machine learning model capable of predicting student academic outcomes using educational and engagement-related data.

A Random Forest Classifier was selected, trained, evaluated, and tuned using GridSearchCV. The final model achieved an accuracy of approximately 68.26% while identifying assessment activity and learning engagement as the most influential factors affecting student success.

The results demonstrate that machine learning can effectively support early identification of at-risk students and provide valuable insights into learning behavior. The completed model forms the foundation for future enhancements, including advanced algorithms, dropout risk prediction, real-time monitoring, and educational intervention systems.